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Lesson notes

Quadratic Equations: Lesson Notes

SS 1 Mathematics. Five short lessons with worked examples, exercises and answers.

Subject	Class	Topic	Lessons
Mathematics	**SS 1**	**Quadratic equations**	**5**

1. What makes an equation quadratic

A quadratic equation has the unknown raised to the power 2 as its highest power. We write it in standard form: $\mathbf{ax^2 + bx + c = 0}$, where a is not 0.

Examples: $x^2 - 5x + 6 = 0$ (a = 1, b = −5, c = 6); $2x^2 + 3x = 9$ becomes $2x^2 + 3x - 9 = 0$; $x^2 = 49$ becomes $x^2 - 49 = 0$.

Every quadratic has at most two solutions, called its **roots**.

2. Solving by factorisation

If two numbers multiply to give zero, at least one of them is zero. This is the **zero product rule**.

For $x^2 + bx + c = 0$, find two numbers that multiply to c and add to b, write the equation as $(x + p)(x + q) = 0$, then set each bracket to zero.

Worked example
Solve $x^2 - 5x + 6 = 0$. The numbers −2 and −3 multiply to 6 and add to −5. So $(x - 2)(x - 3) = 0$, giving **x = 2 or x = 3**.

3. Completing the square

Use this method when the quadratic does not factorise neatly. Move c to the right, add $(b \div 2)^2$ to both sides, write the left side as a perfect square and take square roots (remember ±).

Worked example
Solve $x^2 + 6x - 7 = 0$. $x^2 + 6x = 7$; add 9 to both sides: $(x + 3)^2 = 16$; $x + 3 = \pm 4$, so **x = 1 or x = −7**.

4. The quadratic formula

For $ax^2 + bx + c = 0$: $\mathbf{x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a}$.

The **discriminant** $b^2 - 4ac$ tells you what to expect: positive gives two real roots, zero gives one repeated root, negative gives no real roots.

Worked example
Solve $2x^2 - 4x - 3 = 0$. Here $a = 2$, $b = -4$, $c = -3$, so $b^2 - 4ac = 16 + 24 = 40$ and $x = (4 \pm \sqrt{40}) \div 4$, giving **$x \approx 2.58$ or $x \approx -0.58$**.

5. Word problems

Let the unknown be x, write an equation from the story, solve it, then reject answers that make no sense, such as negative lengths, ages or prices.

Worked example
A garden is 3 m longer than it is wide and its area is 40 m². $x(x + 3) = 40$ gives $x^2 + 3x - 40 = 0$, so $(x + 8)(x - 5) = 0$. The width is **5 m** and the length is 8 m.

Exercises

1. Write $3x^2 = 7x - 2$ in standard form and state a, b and c.

2. Solve $x^2 - 9x + 20 = 0$ by factorisation.

3. Solve $2x^2 + 7x + 3 = 0$.

4. Solve $x^2 + 4x - 5 = 0$ by completing the square.

5. Find the discriminant of $x^2 + 2x + 5 = 0$ and say what it tells you.

6. Two consecutive whole numbers multiply to 156. Find them.

Answers

1. $3x^2 - 7x + 2 = 0$; $a = 3$, $b = -7$, $c = 2$.

2. $(x - 4)(x - 5) = 0$, so $x = 4$ or $x = 5$.

3. $(2x + 1)(x + 3) = 0$, so $x = -1/2$ or $x = -3$.

4. $(x + 2)^2 = 9$, so $x = 1$ or $x = -5$.

5. $4 - 20 = -16$, which is negative, so there are no real roots.

6. $x(x + 1) = 156$ gives $x = 12$, so the numbers are 12 and 13.